

UCF Physics: AST 5765/4762: (Advanced) Astronomical Data Analysis
Homework 6: Median Combination
Due Wednesday, October 7, 2026, 2:30 pm

Remember, at all points, to explain what you do in your log and keep appropriate notes in your code. Not doing so can result in lost points!

Please read the Preamble in `Practicum_4.pdf`.

1. **(5 points)** Use your terminal to make a new folder `hw7_<yourname>` under your `ast4762/` homework folder. For this homework, your main homework file is a Python file named `hw6_<username>.py` (or `hw6_<username>.ipynb`, but remember to save it as a `.py` file before submission). Save it in your homework folder. Remember to commit your files and push to GitHub. Start the file with the usual header.

2. **(30 points)** As we have seen in class, the median-combination method combines a stack of images into a single image by taking the median of the image values at each pixel location (x, y) . If you have a stack of 5 images, with each image having 1000×1000 pixels, you take 1,000,000 medians of five elements each, to make a single 1000×1000 image. This reduces noise and eliminates *most* of the bad pixels in your data.

a) Find an appropriate function that allows you to median-combine the images in a single function call, without loops. The function should accept your 3D array and return the median-combined 2D array. Name the function and state anything special you need to do to get this behavior.

b) Call the function on your dark data. Print the value of pixel index `[217, 184]` of your median-combined dark image.

c) Add a `HISTORY` entry to your dark header, stating that it is a median-combined dark frame.

d) Write the median dark (with the modified header from c) to a file named `dark_13s_med.fits`.

e) Subtract the median-combined dark frame from *each* object frame. Use no loops. Write **only** the first frame to a file named `hw7_<yourname>_prob2_graph1.fits`. In your main homework file, print the value of pixel index `[217, 184]` of the first object frame (Python index 0) before and after dark subtraction.

4. **(5 points)** Test that your code works on Stokes and submit your homework. See prior assignments for details.